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HANDELSHØJSKOLEN

TEXT CLASSIFICATION

Bonus Material - Naive Bayes Classification

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Topics

Handling Imbalanced Data

K-fold Cross-Validation

Improving Model Performance

Laplace smoothing

How to avoid Underflow?

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Handling Imbalanced Data

Resampling

- Oversampling the minority class
- Undersampling the majority class
- SMOTE (Synthetic Minority Over-sampling Technique)

Ensemble Methods

(Bagging and Boosting)

- Bagging - training multiple classifiers on different subsets of the data and then aggregating their predictions
- Boosting - training multiple classifiers sequentially, each trying to correct the errors of the previous one

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Handling Imbalanced Data

Adjusting Class Prior (Weights)	• Assign higher weights to the minority class during training to make the model more sensitive to it. • Requires cross-validation to find the optimal priors.
Cost-sensitive Learning	• Modify the learning algorithm to prioritize reducing misclassifications of the minority class • Penalizing wrong classifications more heavily for the minority class to make the model more sensitive to it.

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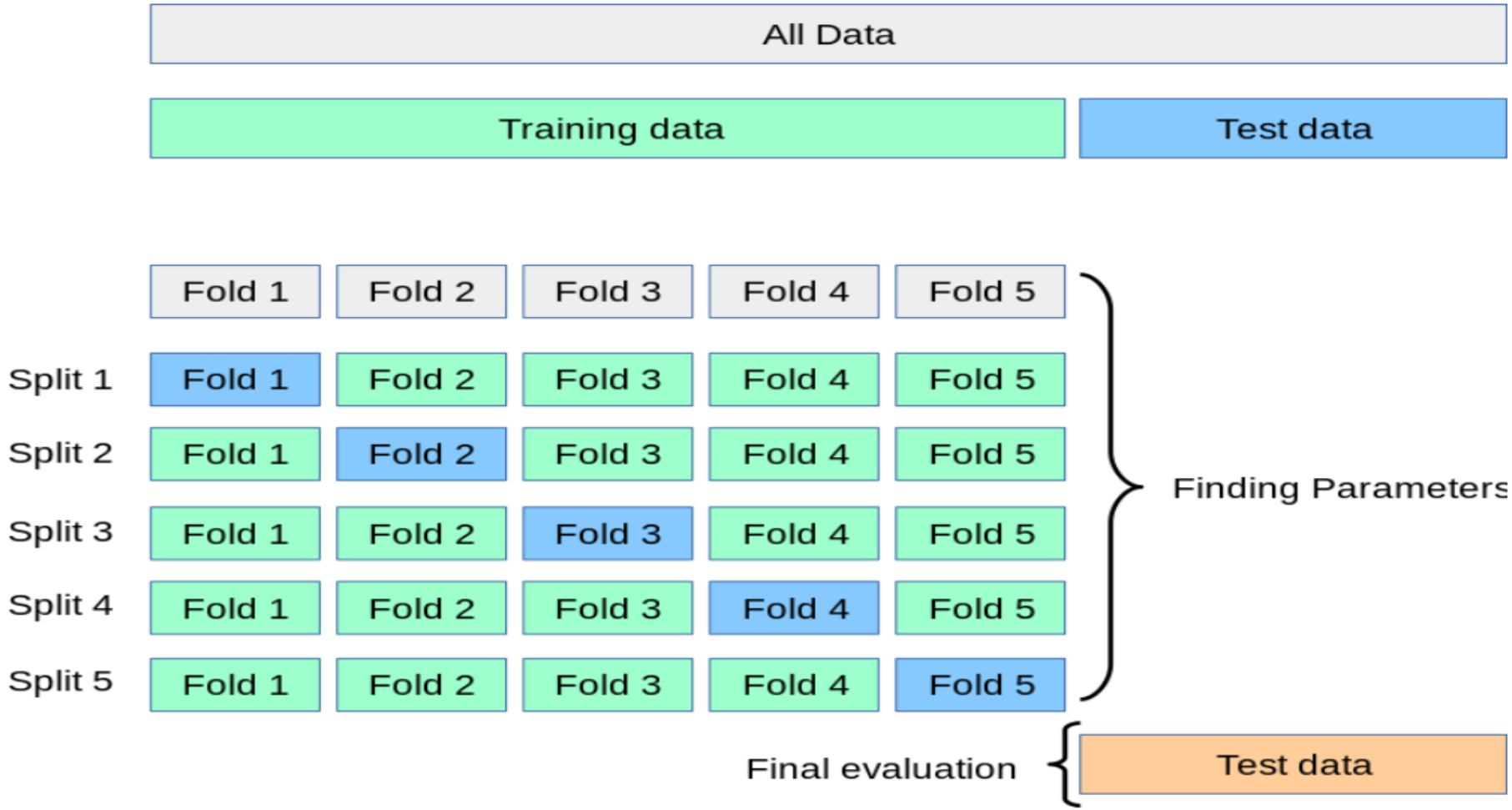
K-fold Cross-Validation



"Train and test the model multiple times on different data splits."

Why Cross-Validation?

"Train and test the model multiple times on different data splits."

Model Evaluation

- provides a more accurate measure of a model's predictive performance.

Generalization

- helps understanding how well the model generalizes to unseen data.

Hyperparameter Tuning

- essential for selecting the best hyperparameters, such as the smoothing parameter.

Reducing Bias and Variance

- helps in identifying the balance between bias and variance

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How to improve performance?

Feature Selection	• Identify and retain only the most informative features and reduce dimensionality.
Term Frequency-Inverse Document Frequency	• Use TF-IDF instead of BoW to give more weight to terms that are important and unique to a document but less weight to common terms.
Hyperparameter Tuning	• Optimize parameters like the smoothing factor (alpha) • Utilize methods like grid search or random search
Ensemble Techniques:	• Combine multiple Naive Bayes classifiers with methods like bagging or boosting to reduce variance and bias

Feature Extraction: TF-IDF

*TF-IDF = TF * IDF*

- A statistical measure used to evaluate the importance of a word in a document relative to a collection of documents (corpus).
- TF (Term Frequency):
  - Measures how frequently a term occurs in a document.
  - Calculated as the number of times a word appears in a document divided by the total number of words in that document.
- IDF (Inverse Document Frequency):
- Measures the importance of the word across the corpus.
- Calculated as the logarithm of the number of documents in the corpus divided by the number of documents containing the word.

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Bag of Words vs. TF-IDF

Feature	Bag of Words	TF-IDF
Concept	Counts word frequencies.	Weighs frequencies by inverse document occurrence.
Dimensionality	High, can lead to sparse matrices.	High, but weights may reduce feature noise.
Complexity	Simple, easy to implement.	Slightly more complex due to IDF calculation.
Word Importance	Treats all words equally.	Differentiates words based on document uniqueness.
Context Sensitivity	None, ignores context and word order.	Some, by de-emphasizing common words.
Performance	Effective but may suffer from irrelevance in features.	Often superior, enhances relevance of features.
Use Cases	Presence/absence of words is key.	Distinction between common/unique words matters.
Limitations	High dimensionality, sparsity.	Does not capture word order or semantics.

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Example: Laplace smoothing

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable ~~with~~ no fun

Step 1: Compute Prior

$$\widehat{P}(c_j) = \frac{\#\text{document belong to class } c_j}{\#\text{Total documents}}$$

$$P(\text{pos}) = \frac{2}{5} \text{ and } P(\text{neg}) = \frac{3}{5}$$

Vocabulary = 20

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Example: Laplace smoothing

$$p(w_i|c) = \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$$

	Cat	Documents
Training	-	just plain boring
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	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

Total word in pos class = 9

Total word in neg class = 14

|V|=20

Step 2: Compute Likelihoods

Word	Pos	Neg
predictable	$\frac{0+1}{9+20}$	$\frac{1+1}{14+20}$
no	$\frac{0+1}{9+20}$	$\frac{1+1}{14+20}$
fun	$\frac{1+1}{9+20}$	$\frac{0+1}{14+20}$

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Example: Laplace smoothing

Word	Pos	Neg
predictable	$\frac{0+1}{9+20}$	$\frac{1+1}{14+20}$
no	$\frac{0+1}{9+20}$	$\frac{1+1}{14+20}$
fun	$\frac{1+1}{9+20}$	$\frac{0+1}{14+20}$

$$P(pos) = \frac{2}{5} \quad \text{and} \quad P(neg) = \frac{3}{5}$$

Step 3: Scoring the test sentence

P(pos) . P(S|pos)

$$= \frac{2}{5} \times \frac{1 \times 1 \times 2}{29^3} = 3.2 \times 10^{-5}$$

P(neg) .P(S|neg)

$$= \frac{3}{5} \times \frac{2 \times 2 \times 1}{34^3} = 6.1 \times 10^{-5}$$

Result: Test Sentence belongs to Neg class!

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How to avoid Underflow?

- Taking log doesn't change the ranking of classes!
- The class with highest probability also has highest log probability!
- It's a linear model - Just a max of a sum of weights

$$C_{NB} = \text{argmax}_{c_j \in C} P(c_j) . \prod_{i \in position} P(w_i | c_j)$$

$$C_{NB} = \text{argmax}_{c_j \in C} \log P(c_j) + \sum_{i \in position} \log P(w_i | c_j)$$

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